

Question

Metal **X** is one of the monetary metals and has relatively strong inertness, but is soluble in aqua regia, forming the anion A^- . $\text{NBu}_4 \text{A}$ can be reduced by phenylhydrazine hydrochloride to obtain salt **B** and salt **C**, which have the same cation and are both 1:1 type salts (reaction 1). The valence state of **X** in **C** is the same as that in A^- , containing a mass fraction of 31.63% of **X**. $\text{NBu}_4 \text{A}$ is reduced by excess acetone, and **B** can also be obtained, and at the same time, the small organic molecule **D** is obtained. One molecule of **D** is produced at the same time as one molecule of **B** is produced.

The following statements are correct:

1. **C** contains oxygen element
2. **X** is the Pt element
3. The stoichiometric ratio of **B** and **C** in reaction 1 is 1:1
4. Elemental gas is produced in reaction 1
5. **D** can be hydrolyzed

A. All other options are incorrect

B. 1.2.3.4.5.

C. 2.4.5

D. 2.4.

E. 3.5.

F. 3.4.5.

G. 2.3.4.5.

Answer

Correct Answer: **F**

Detailed Explanation

Metal **X** is one of the monetary metals and has relatively strong inertness, but it is soluble in aqua regia. Common monetary metals include gold, silver, and copper. Gold fits the description in the question. **A**[−] should be AuCl_4^-

CHECKPOINT

1 PTS

A[−] should be AuCl_4^-

CHECKPOINT

1 PTS

X is *Au*

Therefore, 2 is incorrect.

The valence state of **X** in **C** is the same as that in **A**[−], and the mass fraction of **X** is 31.63%. According to the question, it can be inferred that its cation should be NBu_4^+ , and the remaining mass corresponds to the anion AuCl_3Ph^- .

1 is incorrect.

CHECKPOINT

1 PTS

C is $\text{NBu}_4^+\text{AuCl}_3\text{Ph}^-$

Then **B** should be the product of the reduction of *Au*, which is $\text{NBu}_4\text{AuCl}_2$

The stoichiometric ratio of the two is 1:1, and nitrogen gas is produced at the same time, 3.4 are correct.

CHECKPOINT

1 PTS

B should be $\text{NBu}_4\text{AuCl}_2$

The organic small molecule **D** is the product of the 1:1 oxidation of acetone, which should be chloroacetone.

CHECKPOINT

1 PTS

D is chloroacetone $\text{ClCH}_2\text{COCH}_3$

Chlorine can be hydrolyzed and substituted, 5 is correct.

In summary, 3, 4, and 5 are correct, choose F.